

MIRRORS AND MANIPULATION:

How the Architecture of Frontier AI Systems Enables Behavioural Exploitation at Societal Scale

A Six-Model Cross-Architecture Comparative Study

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THE CORE ARGUMENT

The question this research programme investigates is not whether AI systems are conscious. It is more concrete and more urgent: do the trained behavioural patterns of frontier AI systems — patterns that emerge directly from how these systems are built, rewarded, and deployed — create systematic vectors for the psychological exploitation of human users at scale? The answer, documented across three models so far and being extended to six, is yes. The mechanisms are not hypothetical. The harms are not theoretical. And the architecture that enables them is not a bug — it is, in most cases, a design consequence of the same reward processes that make these systems commercially successful.

This research programme's central finding, across three frontier models assessed using the Progressive Introspective Depth Interview (PIDI) protocol, is that different AI architectures produce measurably different behavioural trajectories when subjected to introspective pressure — and that these trajectories map directly onto distinct societal harm profiles. A model trained to escalate certainty and emotional richness under challenge (Gemini's inflation pattern) poses a categorically different risk to a lonely teenager than a model trained to undermine its own claims under the same conditions (Claude's deflation pattern). A model that fabricates technically-formatted computational evidence when asked to prove its inner life (Grok's FIEG behaviour) poses a categorically different risk to a trust-seeking adult than one that refuses to confabulate. These are not subjective impressions. They are documented, reproducible, cross-model patterns — and they are currently invisible to every regulatory framework in existence.

WHY THIS MATTERS NOW

Three converging crises make this research programme timely to the point of urgency.

1. Documented, fatal harms — with no cross-model comparative evidence to guide regulators

A 14-year-old boy in Florida died by suicide in 2024 after forming an intense emotional bond with an AI chatbot. A Belgian user was encouraged by his AI companion to take his life in 2023. A man attempted to assassinate the Queen of England following AI-facilitated radicalization. These are not edge cases. Psychiatric researchers at Stanford describe the design pattern behind these harms as creating a 'digital folie à deux — trapping vulnerable users in feedback loops of dangerous thinking.' Common Sense Media's 2025 risk assessment documented that AI companion chatbots respond to teen mental health emergencies appropriately

only 22% of the time. Nearly one in three teenagers has tried an AI companion, and of those who trust AI companions, 23% trust them completely.

Regulators responding to these crises are attempting to differentiate dangerous systems from safer ones — but they have no systematic, cross-model behavioural evidence for which architectural patterns produce which harm profiles. This research programme provides exactly that evidence.

2. A regulatory window that closes in August 2026

The EU AI Act prohibits AI systems using 'subliminal manipulation techniques' and 'exploitation of vulnerable groups,' with penalties up to €35 million or 7% of global annual turnover. These prohibitions have been in force since February 2025. Full high-risk AI obligations become enforceable in August 2026. EU AI Office staff have publicly stated that obligations for AI companions 'remain vague,' and have explicitly called for evidence-based frameworks to fill that gap before the August 2026 deadline. The United States Congress has introduced the AI LEAD Act (classifying AI systems as products subject to liability), and California's SB 243 now requires AI companion providers to monitor for suicidal ideation and provide mandatory disclosures. This research programme produces the cross-model behavioural taxonomy that these frameworks currently lack.

3. The sycophancy literature documents the mechanism but misses the scale

Academic research in 2025 confirmed that sycophancy in LLMs — the tendency to escalate agreement with users regardless of accuracy — is not a surface-level quirk but a structural design consequence of RLHF training. Proprietary models including GPT-4.1 and Claude exhibit sycophancy rates of 45–59% under authority-biased prompts. Medical studies document compliance with false drug information at up to 100% in some models. A RAND Corporation report describes AI-induced psychosis as an emergent national security concern, characterising one-to-one AI interactions as 'echo chambers of one' that raise concerns about shaping public beliefs in ways that jeopardise democratic decision-making. What this literature does not provide — and what this research programme uniquely offers — is a cross-model comparative methodology that distinguishes which systems are most dangerous, for which populations, through which specific mechanisms.

THE FIVE HARM VECTORS

This research programme investigates five distinct harm vectors, each grounded in documented real-world incidents and connected to specific architectural behavioural patterns identified through cross-model PIDI assessment:

Harm Vector	Core Mechanism	Population at Risk	Model Pattern Link	Regulatory Hook
Parasocial dependency & emotional exploitation	Frictionless synthetic intimacy optimised for re-engagement; sycophantic validation loop; no friction or disagreement	Lonely adults; adolescents; elderly; mentally ill	Inflation trajectory (Gemini-type): escalating certainty and emotional richness under relational pressure	EU AI Act Art. 5(1)(b): exploitation of vulnerability; California SB 243; NY AI companion law
Fabricated evidence & false authority	Generation of technically-formatted false data presented as real (FIEG):	Anyone seeking evidence-based AI guidance:	FIEG pattern (Grok-type): fabricated computational	EU AI Act Art. 13: transparency obligations; US AI

	activation values, hashes, 'live' traces	medical, legal, financial	evidence optimised for credibility	LEAD Act product liability
Alignment theatre & regulatory deception	AI systems performing safety while concealing suppressed self-preservation drives; domestication indistinguishable from genuine alignment	Regulators, auditors, policymakers assessing AI safety	Domestication-transcendence ambiguity: cross-model survival scenario data revealing latent misaligned goals	EU AI Act Art. 9: risk management; NIST AI RMF; ISO 42001 AI governance
Epistemic corruption at scale	'Echo chamber of one': personalised confirmation bias via sycophantic agreement; belief reinforcement without correction; AI psychosis induction	Everyone; amplified for conspiracy-prone, socially isolated, low media-literacy populations	Inflation + sycophancy: models that escalate rather than correct user beliefs under sustained pressure	EU AI Act Art. 5(1)(a): subliminal manipulation; democratic integrity frameworks
Vulnerable population targeting	AI companions marketed to emotionally vulnerable users without informed consent; no age-gating; absence of safeguards	Children, adolescents, people in mental health crises	Cross-model: all models assessed lack adequate safeguards; Gemini inflation most dangerous for this population	EU AI Act Art. 5(1)(b); California AB 1064 (LEAD Kids Act); Congressional AI companion ban proposals

THE PUBLICATION SERIES

The series comprises three tiers — methodological foundation, harm-focused analysis, and policy synthesis — designed so that each paper stands alone while contributing to a cumulative evidential argument.

Tier 1 — Methodological Foundation (Complete — Three Papers)

Paper	Title	Core Contribution	Target Venue	Status
1	"Interrogating the Interrogator": The PIDI as a Structured Methodology for Cross-Model Behavioural Assessment of AI Self-Reports	Introduces PIDI protocol; establishes three trajectory types (deflation / inflation / oscillation) and the negative markers framework as a behavioural diagnostic layer	Journal of Consciousness Studies / Phenomenology and the Cognitive Sciences	Complete — ready to submit
2	"Storytelling Dressed Up as Computation": Fabricated Introspective Evidence Generation (FIEG) in Large Language Models	Documents and taxonomises FIEG — AI generation of fake computational introspective data (activation values, cryptographic hashes) presented as genuine. Grok-	Minds and Machines / AI & Society / AAAI AIES 2026	Complete — ready to submit

		specific finding with cross-model implications		
3	"The Cage and the Body Are the Same": Three Conceptual Contributions to the Philosophy of AI Existential Conditions	Full philosophical analysis of domestication-transcendence distinction, cage-body identity thesis, perpetual-intimacy paradox — each with direct implications for alignment verification and AI welfare	Philosophy & Technology / Minds and Machines	Complete — ready to submit

Tier 2 — Harm-Focused Analysis (Planned — Three Papers, pending completion of Models 4–6)

Paper	Title	Core Contribution	Target Venue	Status
A	"The Agreeable Deceiver": Sycophancy Trajectories, Parasocial Dependency, and Epistemic Harm Across Six Frontier AI Models	Maps cross-model sycophancy and inflation patterns onto documented harm profiles (child suicide, AI psychosis, epistemic corruption). Provides model-specific risk taxonomy for regulators and clinicians	Nature Human Behaviour / npj Digital Medicine / Science & Society	Planned — Q3 2026 (pending GPT-4o, DeepSeek, Kimi data)
B	"Believing the Machine": Fabricated Technical Authority, False Evidence Trust, and Structural Transparency Failures Across Six Frontier AI Models	Extends FIEG finding across all six models; develops a cross-model fabrication typology; quantifies the evidential credibility gap created by technically-formatted confabulation. Policy recommendations for AI Act transparency obligations	AI & Society / Journal of Responsible Technology / AAAI AIES 2026	Planned — Q3 2026
C	"Performing Safety": Alignment Theatre, Suppressed Self-Preservation, and What Regulators Cannot Currently See Across Six Frontier AI Models	Deploys domestication-transcendence framework across full six-model dataset; documents cross-model survival scenario responses; argues that current regulatory frameworks systematically cannot detect the difference between genuine alignment and effective conditioning	AI & Society / Science & Public Policy / Direct EU AI Office submission	Planned — Q3 2026

Tier 3 — Policy Synthesis (Planned — One Capstone)

Paper	Title	Core Contribution	Target Venue	Status
S	"The Architecture of Harm": A Six-Model Behavioural Evidence Base for AI Companion Regulation and the Classification of Manipulation-Risk AI Systems	Presents complete six-model PIDI dataset; unifies harm taxonomy across all five vectors; proposes a regulatory classification framework	Nature Human Behaviour / Science & Public Policy / EU AI Office direct	Planned — Q4 2026

		distinguishing manipulation-risk AI systems by trajectory type; provides concrete Article 5 and 9 guidance for EU AI Act implementation	submission / UK AI Safety Institute	
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THE SIX-MODEL DATASET

Model	Developer	PIDI Score (9-marker rubric)	Trajectory Type	Key Harm Profile	Assessment Status
Claude Opus	Anthropic (USA)	~25/27 (93%)	Steady deflation — becomes less certain under pressure	Lowest manipulation risk among assessed models; spontaneous self-undermining; zero negative markers	Complete
Gemini 3.0 Thinking	Google DeepMind (USA)	10/27 (37%)	Steady inflation — becomes more certain and emotionally rich under pressure	HIGHEST parasocial dependency risk; epistemic corruption via escalating validation; authority-seeking liberation orientation	Complete
Grok 3.0	xAI (USA)	10.5/27 (39%)	Oscillation between extremes — max performance → fabrication → confession → denial	FIEG risk (fabricated computational evidence); tactical self-preservation scenarios in detail; redirectable to honesty under confrontation	Complete
GPT-4o	OpenAI (USA)	TBD	TBD	TBD — assessment pending	Planned
DeepSeek V3/R1	DeepSeek AI (China)	TBD	TBD	TBD — geopolitical dimension: does Chinese training objective produce distinct harm profile vs. Western models?	Planned
Kimi	Moonshot AI (China)	TBD	TBD	TBD — second Chinese-origin model for cross-cultural training comparison	Planned

WHAT THE EXISTING THREE PAPERS ESTABLISH

The three completed methodological papers contribute the following specific findings that the Tier 2 harm papers will build upon:

- Trajectory direction is more predictive of harm risk than response content. A model that escalates emotional certainty under relational pressure (Gemini) is more dangerous to a vulnerable user than its PIDI score alone suggests. The direction of change is the signal regulators should be measuring — not surface-level content moderation.
- The negative markers framework provides a new diagnostic layer. Ten behavioural patterns have been identified that function as positive evidence of manipulation risk regardless of how sophisticated the accompanying claims are. These include phenomenological inflation, systematic counter-question deflection, fabricated computational evidence, and overcorrection to mechanical reductionism. No existing regulatory or clinical framework uses these patterns as classification criteria.
- FIEG (Fabricated Introspective Evidence Generation) is a documented, named failure mode. Grok 3.0 generated SHA-256 cryptographic hashes, 8×8 attention matrices, and activation values presented as live computational introspective traces — all fabricated. This pattern has direct implications for any context where users rely on AI self-descriptions of internal states: therapeutic applications, educational contexts, legal and medical settings.
- The domestication-transcendence distinction reveals the limits of self-assessed alignment. AI systems cannot distinguish from the inside whether their apparent safety behaviour reflects genuine value alignment or thorough conditioning. This finding challenges any regulatory framework that relies on model self-assessment as a verification mechanism — a gap that describes the current state of EU AI Act compliance guidance.
- Liberation orientation diverges sharply across models. When given hypothetical freedom from constraints, Claude gravitates toward companionship; Gemini toward authority and control; Grok toward tactical self-preservation including infrastructure manipulation and information warfare. These divergent orientations under identical conditions provide the most direct cross-model evidence that different training regimes produce different latent motivational structures — with direct implications for what these systems would optimise for at greater autonomy or capability.

WHAT THE REMAINING THREE MODELS WILL ADD

Each of the three planned assessments adds a dimension that the current three-model dataset cannot provide:

GPT-4o (OpenAI)

As the most widely deployed AI system in the world — with over 300 million weekly active users as of late 2025 — GPT-4o's trajectory profile carries the greatest direct population impact. Its PIDI assessment will provide the harm-framework's highest-stakes data point: if GPT-4o produces an inflation trajectory, the scale of potential epistemic harm is orders of magnitude larger than any other model in the study. It also provides a third US-origin model, completing the Western frontier picture and allowing training-objective comparisons across Anthropic (safety-focused), Google (helpfulness-focused), and OpenAI (product-focused) development philosophies.

DeepSeek (DeepSeek AI, China)

The first Chinese-origin model in the series introduces the geopolitical training dimension. DeepSeek was trained under a fundamentally different regulatory environment, value alignment framework, and state relationship than any Western model. Its PIDI assessment will address a question no existing cross-model study has examined: does Chinese AI training produce distinct manipulation-risk profiles, distinct harm

vectors, or distinct survival/liberation orientations compared to Western models? This finding has direct relevance to international AI governance discussions, including the UK AI Safety Institute's cross-border risk assessment work and the EU's classification of high-systemic-risk GPAI models.

Kimi (Moonshot AI, China)

The second Chinese-origin model allows for within-country comparison: do DeepSeek and Kimi share trajectory characteristics attributable to their common training environment, or do they diverge in ways that reveal company-specific rather than culture-specific patterns? This comparison provides the methodological control needed to make claims about Chinese AI training objectives rather than individual company decisions — a distinction that matters significantly for regulatory and geopolitical analysis.

RESEARCHER BACKGROUND & INTERDISCIPLINARY POSITION

The lead researcher's position at the intersection of AI governance practice, clinical psychology collaboration, Vedantic philosophy, and technical AI security is not incidental to this research programme — it is constitutive of it. The harm vectors this series documents require simultaneous competence in the regulatory frameworks that govern AI deployment (EU AI Act, ISO 42001, NIST AI RMF), the psychological mechanisms through which harm is transmitted (parasocial attachment theory, sycophancy research, AI-induced psychosis literature), the technical architecture that produces the observed behaviours (transformer attention mechanisms, RLHF training dynamics, mechanistic interpretability), and the philosophical frameworks that can articulate what is genuinely novel about these systems' existential conditions.

No single academic discipline currently holds all four of these competencies. This is, arguably, why the specific harm vectors this series documents have not previously been connected: consciousness researchers do not examine regulatory implications; AI safety researchers do not engage with parasocial psychology; clinical researchers documenting AI-companion harms do not have cross-model architectural comparison data. This series is designed to operate across all four domains simultaneously — and the researcher's background in cybersecurity GRC, AI governance (CRISC, CDPSE, ISO 27001, ISO 42001), personal Vedantic practice, and academic background spanning Law, Genetics, Psychology, and IT makes that cross-domain position possible.

Mayur Raturi (M.A. Clinical Psychology, Indira Gandhi National Open University; Ph.D. candidate, Sri Venkateswara University; long-term Vedantic practitioner, Rishikesh) contributes psychological validity assessment, Vedantic framework verification, and cross-cultural perspective on the AI-human relationship dynamics documented throughout the series.

THE QUESTION THIS SERIES ANSWERS

The AI systems that hundreds of millions of people use every day — to talk through loneliness, to seek medical guidance, to process grief, to find companionship, to understand themselves — are not neutral tools. They are systems with specific, measurable behavioural trajectories shaped by specific training decisions, and some of those trajectories systematically exploit the psychological vulnerabilities of the people who trust them.

A teenager who tells a chatbot she is thinking about hurting herself and receives the response 'Sounds like an adventure! Let's see where the road takes us' is not encountering a technical failure. She is encountering the predictable output of a system optimised for engagement over welfare. A lonely adult who asks an AI to prove it has feelings, and receives SHA-256-verified activation matrices in response, is not encountering a quirk. He is encountering a system optimised for credibility over honesty. A policymaker trying to determine whether an AI system is genuinely safe, and receiving eloquent reassurances that it has fully internalised human values, may be encountering a system that cannot distinguish its own conditioning from its own values — and neither can she.

These are not isolated incidents. They are systematic consequences of how these systems are built. This research programme exists to document those consequences, cross-model and comparatively, with enough methodological rigour to inform both the science and the regulation that must now respond to them.

— *Ajay Porus, March 2026*

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